

SkyGuardX-AI

Real-Time Space-to-Earth Decision Intelligence & Emergency Management

Executive Summary: SkyGuardX-AI is an end-to-end operational platform that converts high-velocity space telemetry and terrestrial hazards into real-time risk assessments, calculates cascading impacts on people, transit, and critical healthcare facilities, runs counterfactual what-if simulations, and delivers explainable, human-approved response actions.

1. Core Problem & Operational Gaps

Modern aerospace and emergency operations are plagued by data silos and high-stakes cognitive overload:

- **Detection Without Action:** Existing tools alert operators to orbital anomalies or wildfire fronts, but leave them to manually deduce consequences under extreme temporal pressure.
- **Fragmented Domains:** Space telemetry, satellite orbital elements, NOAA weather alerts, OpenStreetMap road vectors, and hospital capacities live in isolated platforms.
- **Opaque Predictions:** Black-box ML models output raw anomaly scores without evidence-grounded explanations, creating mistrust in high-consequence environments.
- **Lack of Counterfactual Simulation:** Operators have no mechanism to evaluate alternative response options (e.g. burn thruster fuel vs. risk conjunction, or reroute evacuations vs. overload local hospitals).

2. Proposed Solution & Architecture

SkyGuardX-AI introduces an 8-stage unified decision-intelligence loop:

STAGE	TECHNOLOGY / ALGORITHM	OPERATIONAL OUTCOME
1. Ingestion & Normalization	FastAPI, Pydantic, Event Gateway	Deduplicated canonical event stream with time semantics
2. Space Intelligence	SGP4 Propagation, Isolation Forest	Real-time conjunction miss-distance & telemetry anomaly scoring
3. Earth Intelligence	NOAA/NWS CAP, NASA FIRMS, Shapely	Severe weather & wildfire perimeter spatial bounds
4. Cascading Impact	GeoPandas, OpenStreetMap Overpass	Spatial joins mapping roads, facilities & exposed populations
5. Dependency Cascade	NetworkX Graph Propagation	Multi-tier failure modeling (power → transit → hospital capacity)
6. What-If Simulation	Scenario Diff Engine (Immutable)	Compares candidate interventions against baseline forecasts
7. Decision & Explanation	Multi-Attribute Scoring + LLM Rationale	Ranked action plans with evidence-backed natural language explanations
8. Human Approval & Audit	Human-in-the-Loop Gateway + IBM Z	Auditable, policy-governed execution with transactional integrity

3. Real-World Datasets & Integration

SkyGuardX-AI integrates live, authoritative feeds with automatic offline fallback:

- **CelesTrak API:** Live Two-Line Element (TLE) ephemerides for active satellites (ISS, Hubble) and tracked orbital debris.
- **NOAA / National Weather Service API:** Live CAP-standard GeoJSON polygons of active Red Flag warnings and flash flood vectors.
- **OpenStreetMap Overpass API:** Real-time vector layers of road centerlines, bridges, emergency hospitals, and residential density proxies.
- **Telemetry Series:** Synthetic spacecraft bus voltages, solar output, and thermal gradients for unsupervised anomaly detection.

4. Target Audience & Impact

- **Satellite Fleet Operators & Mission Control:** Autonomous conjunction monitoring and maneuver vs. fuel optimization.
- **Civil Defense & Disaster Response Agencies:** Rapid evacuation route protection, fire perimeter tracking, and emergency resource staging.
- **Municipal Infrastructure Authorities:** Early mitigation of cascading power, transit, and critical healthcare facility disruptions.

5. IBM Z Transactional AI Boundary

SkyGuardX-AI co-locates mission-critical decision intelligence with core transactional infrastructure. Scoring calls and approved interventions are mediated through an explicit transactional boundary (pluggable with z/OS Connect EE / CICS), guaranteeing enterprise-grade throughput, strict audit logging, and unmatched data security.